Foundation III: The Perfuming Loop
A Conditional One-Step Source-Level Feedback Program

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Abstract

Foundation I [1] constructs a conditional single-source chain from a source datum S_{source} to an admissible manifest pair $(\mathcal{A}_c, \omega)$. Foundation II [2] extends this to two sources, isolating a shared manifest sector. Both chains are one-directional: source data produce manifest sectors, but the manifest sectors do not feed back to the source. The present paper, Foundation III, addresses the return arrow.

T-DOME III [6] supplies a calibration loop at the ego level under explicit residual-likelihood, convexity, drift, and hardware assumptions: the observer's prediction-error residuals drive a natural-gradient update of the reference-frame parameter σ, with tracking controlled by a local Lyapunov estimate. Paper III of the Consciousness Trilogy [5] now treats the former drift-floor theorem as withdrawn and leaves only a supplied arithmetic comparison score for finite self-models. Neither construction closes the loop at the source level: the ego adjusts its frame, but the decoherence semigroup $\{\mathcal{E}_t\}$ that generated the manifest sector in the first place is left untouched.

Foundation III defines a *source-level feedback map* that lifts the ego-level calibration signal to a controlled perturbation of the decoherence semigroup. Under named regularity and coupling conditions, we prove a conditional one-step closed-loop foothold with three components. First, the feedback-modified semigroup remains a valid source datum in the pointwise σ-weak Heisenberg-picture sense used by Foundation I v2. Second, the supplied self-description comparison functional under the closed loop satisfies a modified algebraic comparison bound that depends on the feedback coupling strength. Third, the one-directional Foundation I chain upgrades only to a conditional one-step closed loop:

$$\begin{aligned} S_{\text{source}} \longrightarrow \mathcal{O}_B \longrightarrow \text{calibration signal} \longrightarrow S_{\text{source}}[\text{updated}] \\ \longrightarrow \mathcal{O}_B[\text{updated}] \longrightarrow (\mathcal{A}_c, \omega)[\text{updated}]. \end{aligned}$$

The paper does not claim that the loop eliminates the self-description deficit. It proves only that the loop is well-posed and that the deficit transforms in a controlled way. The question of whether iterated perfuming converges to a fixed point, and what such a fixed point would mean structurally, is left as an open problem.

Contents

1 Introduction **3**

2 Feedback Lift and Regularity Conditions **4**

3 Source-Datum Preservation **7**

4 Modified Local Comparison Functional **9**

5 Conditionally Well-Posed One-Step Return Map **11**

6 Open Problems **12**

7 Conditional Input Ledger **13**

A Imported Interfaces **14**

References **16**

1 Introduction

Foundation I [1] solved the first source-side bridge in the Consciousness Series. Starting from a source datum

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0}),$$

it isolated a conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega),$$

where $\mathcal{O}_A$ is a persistent projected record process, $\mathcal{O}_B$ is a foreground frame available under the explicit conditional coding interface of Foundation I, and $(\mathcal{A}_c, \omega)$ is the manifest accessible pair. The chain is one-directional: source data produce a manifest sector, and the series then hands that sector to Paper I of the Consciousness Trilogy [3] for the geometry–modular-flow theorem and to Paper II [4] for the higher-dimensional consistency analysis.

Foundation II [2] extended that source-side analysis to the first multi-source setting. There the main result was again one-directional. Given two source chains sharing one ambient Type III_1 substrate, one obtains a conditional inclusion theorem for a shared admissible manifest sector. What remained absent in both papers was any formal return arrow from manifest calibration back to the source dynamics that generated the manifest sector in the first place. Foundation II enters the present paper only as programme boundary calibration for the later multi-source feedback problem; no theorem below imports a Foundation II result.

The missing input now comes from two already published directions. T-DOME III [6] supplied a one-step calibration signal at the ego level: prediction-residual statistics determine a natural-gradient update $\delta\sigma$, controlled by a Lyapunov tracking estimate and penalized by the ego-rigidity term $\mathcal{I}_{\text{ego}}$. Paper III of the Consciousness Trilogy [5] proposed a local self-description floor on the manifest side. That reconstruction-loss bound has since been withdrawn: a lossless code for a low-dimensional orbit defeats an n-independent positive floor. Foundation III therefore uses its displayed Fisher expression only as a supplied comparison functional, not as an imported risk or reconstruction theorem. Neither that functional nor the T-DOME calibration model, by itself, modifies the source semigroup $\{\mathcal{E}_t\}$.

The present paper addresses exactly that gap. Its aim is not to prove a global convergence theorem for iterative self-modification or a general iterated-feedback theorem. It does not claim that the self-description deficit is eliminated by feedback, and it does not claim that one can derive a feedback lift from bare multi-scale dynamics without additional structure. Its narrower goal is to identify an honest one-step theorem-level foothold for *source-level feedback from manifest calibration*.

The main design choice is to avoid a formal expression of the type

$$\mathcal{E}_t \mapsto \mathcal{E}_t + \eta\,\Gamma(\delta\sigma, \mathcal{E}_t)$$

as the *definition* of the modified dynamics. Such a first-order expression is useful heuristically, but it does not by itself preserve the semigroup law or complete positivity. Instead, the paper defines a feedback lift through an exact auxiliary semigroup $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ of normal completely positive unital maps and then sets

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t.$$

Once a commutation condition is imposed, the semigroup law is automatic; the first-order feedback map Γ then reappears as the $\eta \downarrow 0$ tangent of this exact construction.

This paper therefore contributes four one-step theorem-level items.

1. We define a *source-level feedback lift* from an admissible calibration signal $\delta\sigma$ to an exact Heisenberg-picture semigroup perturbation on the ambient Type III_1 algebra.

2. We prove a preservation theorem: under named regularity and commutation inputs, the modified family $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ is again a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, so the ambient datum remains a valid source datum.

3. We prove a comparison theorem for a supplied local Fisher functional: if the chosen Foundation I chain is feedback-stable and its parameters remain controlled, then that functional deforms by an explicit small-coupling factor $f(\eta)$ with $f(0) = 1$. No actual deficit bound is inferred.

4. We package the result as a *one-step return datum*, showing that the chain

$$S_{\text{source}} \to \mathcal{O}_B \to \delta\sigma \to S_{\text{source}}^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$$

   is mathematically well-posed under named inputs.

The scope is deliberately narrow. The paper does not solve the constructive origin of the calibration signal, which is imported from T-DOME III. It does not solve the full persistence problem for the Foundation I chain under arbitrary perturbations, which remains a conditional interface. It does not derive explicit optimal coupling windows or fixed points for iterated perfuming. Those are precisely the open problems recorded later.

The paper also keeps the same Type III discipline as Foundation II. No tensor-product factorization is assumed. All dynamics are written in the Heisenberg picture on a common ambient algebra, and continuity means pointwise σ-weak continuity rather than norm continuity. This is not cosmetic: the whole point of the paper is to modify the source semigroup without importing an external finite-dimensional controller.

The paper is organized as follows. Section 2 defines the feedback lift, the regularity package, and the modified source datum. Section 3 proves the source-datum preservation theorem. Section 4 states the extra comparison inputs needed to pass through Foundation I and then proves the comparison-functional theorem. Section 5 packages the resulting one-step return map. Section 6 records the open problems. Section 7 collects the named conditional inputs. Appendix A restates the conditional Foundation I and T-DOME III interfaces and records the withdrawal of the former Paper III floor.

2 Feedback Lift and Regularity Conditions

The first task is to isolate the additional object that Foundation III needs and the exact conditions under which it is allowed to act on the source semigroup.

Definition 2.1 (Admissible calibration signal)**.** Fix a source datum S_{source} together with a chosen Foundation I chain up to Stage B, so that one has

$$\mathcal{O}_B = (\mathcal{P}, \omega_{\mathcal{P},t}, \mathcal{K}_*(t,s), \mathfrak{F}, \pi_{\text{ego}}).$$

An *admissible calibration signal* is a vector

$$\delta\sigma \in T_\sigma \Sigma$$

in the local calibration parameter space of T-DOME III such that:

1. $\delta\sigma$ is obtained from the natural-gradient calibration step imported from T-DOME III;
2. $\sigma + \delta\sigma$ remains inside the Lyapunov neighborhood on which the T-DOME III tracking estimate is valid;
3. there exists a bound $\|\delta\sigma\| \leq R_{\text{cal}}$ for some finite calibration radius $R_{\text{cal}} > 0$.

Remark 2.2 (One-step rather than iterative calibration). Definition 2.1 is intentionally local. The paper treats a single feedback step driven by one calibration signal $\delta\sigma$. An iterated sequence $\delta\sigma_0, \delta\sigma_1, \ldots$ would require a separate fixed point or long-time theory, and nothing in the present paper should be read as supplying that theory.

Definition 2.3 (Feedback regularity package)**.** Let S_{source} be a source datum and let $\delta\sigma$ be an admissible calibration signal. A *feedback regularity package* consists of a family

$$\mathscr{F}^{[\delta\sigma]} = \{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$$

of normal maps on $\mathcal{A}_U$ satisfying the following conditions.

1. **FB-A: CP source lift.** $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, with
$$\mathscr{F}_0^{[\delta\sigma]} = \mathrm{id}_{\mathcal{A}_U}, \qquad \mathscr{F}_r^{[0]} = \mathrm{id}_{\mathcal{A}_U} \quad \forall r \geq 0.$$
2. **FB-B: small-signal differentiability.** For every $A \in \mathcal{A}_U$, the map $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(A)$ is σ-weakly differentiable at $r = 0$. We denote the derivative by
$$\mathfrak{G}_{\delta\sigma}(A) := \frac{d}{dr}\mathscr{F}_r^{[\delta\sigma]}(A)\bigg|_{r=0},$$
and record the normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$, which follows from the unitality in FB-A because $\mathscr{F}_r^{[\delta\sigma]}(\mathbf{1}) = \mathbf{1}$ for all $r \geq 0$.
3. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup:
$$\mathscr{F}_r^{[\delta\sigma]}\mathcal{E}_t = \mathcal{E}_t\mathscr{F}_r^{[\delta\sigma]} \qquad \forall r, t \geq 0.$$
In addition, for the chosen Foundation I chain, the maps $\mathscr{F}_r^{[\delta\sigma]}$ preserve the named structural inputs that are required later to reconstruct Stage A, Stage B, and the manifest pair.

Remark 2.4 (Why the feedback acts through an exact semigroup). The exact feedback semigroup in Definition 2.3 is not an ornament. A first-order perturbation $\mathcal{E}_t + \eta\,\Gamma(\delta\sigma, \mathcal{E}_t)$ does not automatically preserve the semigroup property or complete positivity. The exact semigroup lift is the cleanest way to keep those properties in the Heisenberg-picture Type III setting. The first-order map Γ reappears only after the exact object has been defined.

Remark 2.5 (Meaning of the strict commutation input). The commutation requirement in FB-C is strong and should be read carefully. It does *not* say that feedback is dynamically trivial. For $\eta > 0$,
$$\mathcal{E}_t^{[\eta]} = \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t$$
generally differs from $\mathcal{E}_t$ unless the calibration signal is zero or the coupling is turned off. The role of strict commutation is only to keep the algebra-level time ordering unchanged, so that the modified dynamics remains an exact semigroup without entering a more delicate generator-level perturbation problem. Relaxing FB-C to an approximate-commutation condition is left open.

Proposition 2.6 (A nontrivial commuting-dephasing feedback lift)**.** *The package FB-A–FB-C has a finite-dimensional normal UCP realization that depends nontrivially on $\delta\sigma$ and changes the source dynamics beyond a relabeling.*

Proof. Take $\mathcal{A}_U = M_2 \otimes M_2$, let Z_1, Z_2 be Pauli Z on the two factors, and write $\mathrm{Ad}_{Z_j}(A) = Z_j A Z_j$. Fix $\gamma, \kappa > 0$ and a continuous scalar response

$$q(\delta\sigma) := \|\delta\sigma\|^2.$$

Define bounded generators

$$\mathcal{L}_0 = \gamma(\mathrm{Ad}_{Z_1} - \mathrm{id}), \qquad \mathfrak{G}_{\delta\sigma} = \kappa q(\delta\sigma)(\mathrm{Ad}_{Z_2} - \mathrm{id}),$$

and semigroups

$$\mathcal{E}_t = e^{t\mathcal{L}_0}, \qquad \mathscr{F}_r^{[\delta\sigma]} = e^{r\mathfrak{G}_{\delta\sigma}}.$$

More explicitly,

$$\mathscr{F}_r^{[\delta\sigma]}(A) = a_r A + (1 - a_r) Z_2 A Z_2, \qquad a_r = \frac{1 + e^{-2\kappa q(\delta\sigma) r}}{2}.$$

Thus $\mathscr{F}_r^{[\delta\sigma]}$ is a normal UCP unital semigroup, equals the identity when $\delta\sigma = 0$, and has the stated generator at $r = 0$. The same formula with $(Z_2, \kappa q, r)$ replaced by (Z_1, γ, t) verifies these properties for $\mathcal{E}_t$.

Because Ad_{Z_1} and Ad_{Z_2} commute, so do the two semigroups, which is the algebraic clause of FB-C. A Foundation I input package supported on the joint diagonal algebra is fixed by both dephasings, so the named structural inputs can be chosen feedback-stable in this model. Yet for $A = \mathbf{1} \otimes X$,

$$\mathcal{E}_t(A) = A, \qquad \mathcal{E}_t^{[\eta]}(A) = e^{-2\eta\kappa q(\delta\sigma) t} A.$$

For $\eta > 0$ and $\delta\sigma \neq 0$, the lift therefore adds a real decay channel that the source semigroup did not have; it is not an inert renaming. This is an FD/I witness of compatibility, not a construction of a Type III feedback lift. □

Remark 2.7 (What strict commutation can and cannot add)*.* Proposition 2.6 shows that FB-C does not force the lift to be the identity. It also displays the limitation: the commuting generators are simultaneously diagonal on joint Pauli sectors, so feedback only reweights pre-existing joint modes and does not mix them. Creating new couplings between source sectors generally requires a near-commuting or time-ordered theory, which remains open.

Definition 2.8 (Small-coupling regime and modified semigroup)**.** Fix a feedback regularity package $\mathscr{F}^{[\delta\sigma]}$. A coupling strength $\eta \geq 0$ is in the *small-coupling regime* if $0 \leq \eta < \eta_*$ for some window $\eta_* > 0$ on which the later comparison inputs remain valid. For a first-order interpretation on a return horizon $[0, T]$, we additionally require $\eta T \ll 1$. The exact composed semigroup below exists without this perturbative estimate, but its tangent need not be uniformly small at long times. For such η, define the *feedback-modified semigroup* by

$$\mathcal{E}_t^{[\eta]} := \mathscr{F}_{\eta t}^{[\delta\sigma]} \circ \mathcal{E}_t \qquad (t \geq 0).$$

The corresponding *modified source datum* is

$$S_{\mathrm{source}}^{[\eta]} := (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t \geq 0}).$$

Remark 2.9 (Heisenberg-picture convention). As in Foundation II, all semigroups in Definition 2.8 act on observables. Pointwise continuity therefore means σ-weak continuity on $\mathcal{A}_U$, not norm continuity. Nothing in the present paper uses or claims a norm-continuous generator-level perturbation theorem.

Definition 2.10 (First-order feedback map)**.** Under FB-B, define the *first-order feedback map* by

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) := t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)), \qquad A \in \mathcal{A}_U.$$

Equivalently, for each fixed $t \geq 0$,

$$\mathcal{E}_t^{[\eta]}(A) = \mathcal{E}_t(A) + \eta\, \Gamma(\delta\sigma, \mathcal{E}_t)(A) + o(\eta)$$

in the σ-weak topology as $\eta \downarrow 0$.

Verification of the expansion. By Definition 2.8,

$$\mathcal{E}_t^{[\eta]}(A) = \mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)).$$

Applying the $r = 0$ derivative from FB-B at $r = \eta t$ gives

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A)) = \mathcal{E}_t(A) + \eta t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A)) + o(\eta),$$

which is exactly the stated formula. □

Remark 2.11 (What Definition 2.10 does not prove). The first-order map Γ is merely the tangent of the exact feedback lift. Definition 2.10 does not claim that an arbitrary bounded normal map Γ integrates to a completely positive semigroup, nor does it characterize the most general generator-level perturbations compatible with Type III dynamics. Those questions are left open.

Remark 2.12 (The tangent is not uniform at long times). The identity

$$\Gamma(\delta\sigma, \mathcal{E}_t)(A) = t\, \mathfrak{G}_{\delta\sigma}(\mathcal{E}_t(A))$$

contains an explicit factor of t. Thus the first-order perturbation is controlled for fixed t as $\eta \downarrow 0$, or on windows with $\eta t \ll 1$ under a uniform generator bound. Small η alone does not give a perturbation uniformly small for arbitrarily long times. This is one reason the present result does not iterate to a long-time perfuming theorem.

3 Source-Datum Preservation

We now prove the first of the paper's three main results.

Lemma 3.1 (Semigroup law)**.** *Assume FB-A–FB-C. Then the family* $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ *from Definition 2.8 satisfies*

$$\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]} = \mathcal{E}_{t+s}^{[\eta]} \qquad \forall s, t \geq 0.$$

Proof. For $A \in \mathcal{A}_U$,

$$\begin{aligned}\mathcal{E}_t^{[\eta]}\mathcal{E}_s^{[\eta]}(A) &= \mathscr{F}_{\eta t}^{[\delta\sigma]}\, \mathcal{E}_t\Big(\mathscr{F}_{\eta s}^{[\delta\sigma]}\, \mathcal{E}_s(A)\Big) \\ &= \mathscr{F}_{\eta t}^{[\delta\sigma]}\mathscr{F}_{\eta s}^{[\delta\sigma]}\mathcal{E}_t\mathcal{E}_s(A),\end{aligned}$$

where FB-C is used to commute $\mathcal{E}_t$ past $\mathscr{F}_{\eta s}^{[\delta\sigma]}$. Since $\{\mathscr{F}_r^{[\delta\sigma]}\}$ and $\{\mathcal{E}_t\}$ are both semigroups,

$$\mathscr{F}_{\eta t}^{[\delta\sigma]}\mathscr{F}_{\eta s}^{[\delta\sigma]}\mathcal{E}_t\mathcal{E}_s = \mathscr{F}_{\eta(t+s)}^{[\delta\sigma]}\mathcal{E}_{t+s} = \mathcal{E}_{t+s}^{[\eta]}.$$

□

Lemma 3.2 (Pointwise σ-weak continuity and positivity)**.** *Assume FB-A–FB-C. Then for every $A \in \mathcal{A}_U$,*

$$t \longmapsto \mathcal{E}_t^{[\eta]}(A)$$

is pointwise σ-weakly continuous. Moreover, each $\mathcal{E}_t^{[\eta]}$ is normal, completely positive, and unital.

Proof. Fix $A \in \mathcal{A}_U$ and a normal functional $\varphi \in (\mathcal{A}_U)_*$. By Definition 2.8,

$$\varphi(\mathcal{E}_t^{[\eta]}(A)) = \varphi\Big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\Big).$$

The map $t \mapsto \mathcal{E}_t(A)$ is σ-weakly continuous by the source-datum assumption, and $r \mapsto \mathscr{F}_r^{[\delta\sigma]}(B)$ is σ-weakly continuous for every fixed $B \in \mathcal{A}_U$ by FB-A. To prove continuity of the composition, fix $t_0 \geq 0$ and write

$$\begin{aligned}
&\Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t_0}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\quad\leq \Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\qquad+ \Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big) - \varphi\big(\mathscr{F}_{\eta t_0}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big|.
\end{aligned}$$

For the second term, $\mathcal{E}_{t_0}(A)$ is fixed, so pointwise σ-weak continuity of $\{\mathscr{F}_r^{[\delta\sigma]}\}_{r\geq 0}$ gives convergence to zero as $t \to t_0$.

For the first term, use the predual semigroup $\{\mathscr{F}_{r,*}^{[\delta\sigma]}\}_{r\geq 0}$. Since $\{\mathscr{F}_r^{[\delta\sigma]}\}$ is a pointwise σ-weakly continuous semigroup of normal maps, the predual family is norm-continuous on $(\mathcal{A}_U)_*$. Thus

$$\begin{aligned}
&\Big|\varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_t(A))\big) - \varphi\big(\mathscr{F}_{\eta t}^{[\delta\sigma]}(\mathcal{E}_{t_0}(A))\big)\Big| \\
&\quad= \Big|\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big| \\
&\quad\leq \Big|\big(\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi) - \mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)\big)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big| \\
&\qquad+ \Big|\mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)\big(\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)\big)\Big|.
\end{aligned}$$

The first summand tends to zero because $\mathscr{F}_{\eta t,*}^{[\delta\sigma]}(\varphi) \to \mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)$ in predual norm, while $\mathcal{E}_t(A) - \mathcal{E}_{t_0}(A)$ is norm-bounded by $2\|A\|$. The second summand tends to zero because $\mathcal{E}_t(A) \to \mathcal{E}_{t_0}(A)$ σ-weakly and $\mathscr{F}_{\eta t_0,*}^{[\delta\sigma]}(\varphi)$ is a fixed normal functional. Hence the first term also tends to zero.

Hence $\{\mathcal{E}_t^{[\eta]}\}$ is pointwise σ-weakly continuous.

Normality and complete positivity are preserved under composition of normal completely positive maps, and unitality is preserved under composition of unital maps. Since both $\mathscr{F}_{\eta t}^{[\delta\sigma]}$ and $\mathcal{E}_t$ have those properties, so does $\mathcal{E}_t^{[\eta]}$. □

Theorem 3.3 (Source-datum preservation)**.** *Let*

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$$

be a source datum in the sense imported from Foundation I, let $\delta\sigma$ be an admissible calibration signal, and assume that a feedback regularity package FB-A–FB-C is given. Then for every $\eta \in [0, \eta_*)$, the modified datum

$$S_{\text{source}}^{[\eta]} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t^{[\eta]}\}_{t\geq 0})$$

is again a valid source datum: the family $\{\mathcal{E}_t^{[\eta]}\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$, and Ω remains cyclic and separating for $\mathcal{A}_U$.

Proof. Lemma 3.1 gives the semigroup law and Lemma 3.2 gives pointwise σ-weak continuity, normality, complete positivity, and unitality.

The ambient standard-form triple $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is not changed by the feedback step. In particular, Ω remains cyclic and separating for the same von Neumann algebra $\mathcal{A}_U$. Therefore $S_{\text{source}}^{[\eta]}$ satisfies the same source-datum format as S_{source}. □

Remark 3.4 (What Theorem 3.3 does not prove). The theorem proves only that the feedback-modified dynamics remains in the theorem domain of the Foundation I source-datum definition. It does not prove that the modified datum automatically reproduces the same persistent projection, the same foreground frame, or the same manifest algebra. Those are downstream interface questions and require additional inputs.

Corollary 3.5 (Exact CP lift with first-order tangent)**.** *Under the hypotheses of Theorem 3.3, the map $\Gamma(\delta\sigma, \mathcal{E}_t)$ from Definition 2.10 is the $\eta \downarrow 0$ tangent of an exact Heisenberg-picture completely positive perturbation.*

Proof. This is immediate from Theorem 3.3 together with the first-order expansion of Definition 2.10. □

4 Modified Local Comparison Functional

The preservation theorem keeps the modified dynamics inside the source-datum domain, but it does not yet produce a manifest pair. For that we need the chosen Foundation I chain to survive the perturbation and the supplied local comparison parameters to remain controlled.

Definition 4.1 (Deficit-comparison package)**.** Fix a source datum S_{source}, a chosen Foundation I chain from S_{source} to $(\mathcal{A}_c, \omega)$, and a feedback-modified datum $S_{\text{source}}^{[\eta]}$. We say that *DF-A* holds on the coupling window $[0, \eta_*)$ if:

1. **DF-A(1): feedback-stable Foundation I chain.** For every $\eta \in [0, \eta_*)$, the named conditional inputs used in the chosen Foundation I chain remain available for $S_{\text{source}}^{[\eta]}$. Hence one obtains a modified manifest pair
$$(\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

2. **DF-A(2): Fisher control.** There exist continuous functions $\alpha_F, \beta_{\text{ego}} : [0, \eta_*) \to (0, \infty)$ with
$$\alpha_F(0) = \beta_{\text{ego}}(0) = 1$$
such that
$$\mathcal{I}_F^{[\eta]} \geq \alpha_F(\eta)\, \mathcal{I}_F, \qquad \mathcal{I}_{\text{ego}}^{[\eta]} \leq \beta_{\text{ego}}(\eta)\, \mathcal{I}_{\text{ego}},$$
where $\mathcal{I}_F$ and $\mathcal{I}_{\text{ego}}$ are the baseline manifest-side Fisher sensitivity and ego-rigidity constants, and $\mathcal{I}_F^{[\eta]}$, $\mathcal{I}_{\text{ego}}^{[\eta]}$ are the corresponding modified quantities. Assume also
$$\mathcal{I}_F + \mathcal{I}_{\text{ego}} > 0, \qquad \mathcal{I}_F^{[\eta]} + \mathcal{I}_{\text{ego}}^{[\eta]} > 0,$$
so the comparison functionals are defined.

3. **DF-A(3): supplied local comparison parameters.** A controlled neighborhood is equipped with a stipulated parameter $\rho(\eta) \in [0, \frac{1}{2})$, continuous in η, with baseline value $\rho(0) =: \rho_0$. This parameter and the Fisher quantities define the comparison functional below; they do not import a valid Paper III reconstruction-loss theorem.

Proposition 4.2 (Passage from the modified source datum to the modified manifest pair)**.** *Assume the hypotheses of Theorem 3.3 and DF-A(1). Then for every $\eta \in [0, \eta_*)$, the modified source datum $S_{\text{source}}^{[\eta]}$ admits a conditional Foundation I chain*

$$S_{\text{source}}^{[\eta]} \to \mathcal{O}_A^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

Proof. Theorem 3.3 puts $S_{\text{source}}^{[\eta]}$ inside the theorem domain of the Foundation I source-datum definition. DF-A(1) states exactly that the named conditional inputs used by the chosen Foundation I chain remain available after the feedback perturbation. The Foundation I conditional theorem can therefore be re-run with the modified datum, yielding the stated outputs. □

Definition 4.3 (Baseline and modified local comparison functionals)**.** Under DF-A, define

$$\mathfrak{b}_{\text{loc}} := \frac{(1 - 2\rho_0)\, \mathcal{I}_F}{2\big(\mathcal{I}_F + \mathcal{I}_{\text{ego}}\big)},$$

and

$$\mathfrak{b}_{\text{loc}}^{[\eta]} := \frac{(1 - 2\rho(\eta))\, \mathcal{I}_F^{[\eta]}}{2\big(\mathcal{I}_F^{[\eta]} + \mathcal{I}_{\text{ego}}^{[\eta]}\big)}.$$

These are algebraic comparison scores. The formula is retained from the former Paper III interface, but neither score is asserted here to lower bound reconstruction loss, estimation risk, or a minimax deficit.

Theorem 4.4 (Comparison of supplied local functionals)**.** *Assume the hypotheses of Proposition 4.2 and the full package DF-A(1)–DF-A(3). Then for every $\eta \in [0, \eta_*)$,*

$$\mathfrak{b}_{\text{loc}}^{[\eta]} \geq f(\eta)\, \mathfrak{b}_{\text{loc}},$$

where

$$f(\eta) := \frac{1 - 2\rho(\eta)}{1 - 2\rho_0} \, \frac{\alpha_F(\eta)\big(\mathcal{I}_F + \mathcal{I}_{\text{ego}}\big)}{\alpha_F(\eta)\mathcal{I}_F + \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}}.$$

In particular, $f(0) = 1$, and by continuity $f(\eta) > 0$ for all sufficiently small η.

Proof. By Proposition 4.2, the modified datum yields a manifest pair $(\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$. For fixed $c > 0$ and $d \geq 0$, the function $x \mapsto cx/(x + d)$ is increasing where the denominator is nonzero. Therefore substituting the lower bound for $\mathcal{I}_F^{[\eta]}$ lowers the whole fraction, rather than creating a denominator sign reversal. Likewise the fraction is decreasing in its nonnegative second argument. Using DF-A(2),

$$\mathcal{I}_F^{[\eta]} \geq \alpha_F(\eta)\mathcal{I}_F, \qquad \mathcal{I}_{\text{ego}}^{[\eta]} \leq \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}},$$

so

$$\mathfrak{b}_{\text{loc}}^{[\eta]} \geq \frac{(1 - 2\rho(\eta))\, \alpha_F(\eta)\mathcal{I}_F}{2\big(\alpha_F(\eta)\mathcal{I}_F + \beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}\big)}.$$

Now factor out the baseline score from Definition 4.3:

$$\begin{aligned}\mathfrak{b}_{\text{loc}}^{[\eta]} &\geq \frac{1-2\rho(\eta)}{1-2\rho_0}\,\frac{\alpha_F(\eta)\big(\mathcal{I}_F+\mathcal{I}_{\text{ego}}\big)}{\alpha_F(\eta)\mathcal{I}_F+\beta_{\text{ego}}(\eta)\mathcal{I}_{\text{ego}}}\cdot\frac{(1-2\rho_0)\mathcal{I}_F}{2(\mathcal{I}_F+\mathcal{I}_{\text{ego}})}\\ &= f(\eta)\,\mathfrak{b}_{\text{loc}}.\end{aligned}$$

The formula for $f(0)$ follows from $\rho(0)=\rho_0$ and $\alpha_F(0)=\beta_{\text{ego}}(0)=1$. Continuity then gives positivity for sufficiently small η. □

Remark 4.5 (No monotonicity claim for $f(\eta)$). The theorem asserts only that $f(0)=1$ and that $f(\eta)$ remains positive for sufficiently small coupling. No monotonicity statement is claimed. Depending on the detailed response of the manifest Fisher sensitivity and the ego-rigidity term, one may have $f(\eta)<1$ or $f(\eta)>1$. In particular, a feedback step that increases $\mathcal{I}_F^{[\eta]}$ or decreases $\mathcal{I}_{\text{ego}}^{[\eta]}$ can raise the score rather than lower it.

Remark 4.6 (The former actual-deficit corollary is withdrawn). No saturation assumption can repair the missing premise that $\mathfrak{b}_{\text{loc}}$ is a valid positive lower bound on the actual reconstruction loss. Establishing such a bridge requires either a specified sampling/measurement model and an estimation-risk theorem, or a genuine rate-distortion bottleneck. The present paper supplies neither, so it makes no inequality claim for an actual minimax deficit.

Remark 4.7 (What Theorem 4.4 does not prove). The theorem gives a controlled deformation of a supplied algebraic score. It does not prove that the true minimax deficit is monotone in η, it does not prove that feedback reduces the deficit, and it does not prove that the deficit can be driven to zero. Those stronger claims would require a valid risk/distortion model and a transport theory for optimal self-models.

Remark 4.8 (Why the comparison factor is the right theorem target). The factor $f(\eta)$ isolates exactly what feedback changes at theorem level:

1. the supplied local coefficient $\rho(\eta)$;
2. the manifest Fisher sensitivity $\mathcal{I}_F^{[\eta]}$;
3. the ego-rigidity contribution $\mathcal{I}_{\text{ego}}^{[\eta]}$.

No stronger conclusion is claimed. In particular, the theorem does not say that a better calibration signal necessarily increases $\mathcal{I}_F^{[\eta]}$ or decreases $\mathcal{I}_{\text{ego}}^{[\eta]}$; it only says that if those quantities remain controlled, then the comparison score remains controlled as well.

5 Conditionally Well-Posed One-Step Return Map

We can now state the paper's final theorem.

Definition 5.1 (One-step return datum)**.** Under the hypotheses of Theorem 4.4, the *one-step return datum at coupling* η is the tuple

$$\mathfrak{C}_\eta := \big(S_{\text{source}}, \mathcal{O}_B, (\mathcal{A}_c,\omega), \delta\sigma, S_{\text{source}}^{[\eta]}, \mathcal{O}_B^{[\eta]}, (\mathcal{A}_c^{[\eta]},\omega^{[\eta]})\big),$$

where $\mathcal{O}_B$ and $(\mathcal{A}_c,\omega)$ are the baseline Stage B and Stage C outputs of the chosen Foundation I chain, $\delta\sigma$ is the admissible calibration signal extracted from the retained Stage B data, and $\mathcal{O}_B^{[\eta]}$ together with $(\mathcal{A}_c^{[\eta]},\omega^{[\eta]})$ are the corresponding modified outputs after the feedback step.

Theorem 5.2 (Conditional one-step return-map well-posedness)**.** *Let S_{source} be a source datum, let $\delta\sigma$ be an admissible calibration signal, and assume the packages FB-A–FB-C and DF-A(1)–DF-A(3). Then for every sufficiently small $\eta \in [0, \eta_*)$, and on any fixed return horizon $[0, T]$ with $\eta T \ll 1$, the one-step return datum $\mathfrak{C}_\eta$ of Definition 5.1 is well-defined. Equivalently, the diagram*

$$S_{\mathrm{source}} \longrightarrow \mathcal{O}_B \longrightarrow \delta\sigma \longrightarrow S_{\mathrm{source}}^{[\eta]} \longrightarrow \mathcal{O}_B^{[\eta]} \longrightarrow (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]})$$

is a conditionally well-posed one-step return into the source-datum category.

Proof. The baseline arrow

$$S_{\mathrm{source}} \to \mathcal{O}_B \to (\mathcal{A}_c, \omega)$$

is the chosen Foundation I chain. The arrow

$$\mathcal{O}_B \to \delta\sigma$$

is the imported T-DOME III calibration interface. By Theorem 3.3, the feedback step

$$\delta\sigma \to S_{\mathrm{source}}^{[\eta]}$$

is well-defined at the source-datum level. By Proposition 4.2, the same conditional Foundation I chain can then be run again on $S_{\mathrm{source}}^{[\eta]}$, yielding

$$S_{\mathrm{source}}^{[\eta]} \to \mathcal{O}_B^{[\eta]} \to (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}).$$

Thus every component of $\mathfrak{C}_\eta$ is defined. The supplied comparison score is controlled by Theorem 4.4. This does not turn the one-step return into a self-consistent loop or an actual deficit theorem. □

Corollary 5.3 (Packaged one-step return map)**.** *Under the hypotheses of Theorem 5.2, there is a well-defined map*

$$\mathscr{L}_\eta : \big(S_{\mathrm{source}}, \delta\sigma\big) \longmapsto \big(S_{\mathrm{source}}^{[\eta]}, \mathcal{O}_B^{[\eta]}, (\mathcal{A}_c^{[\eta]}, \omega^{[\eta]}), \mathfrak{b}_{\mathrm{loc}}^{[\eta]}\big).$$

Proof. This is just a repackaging of Theorem 5.2 and Theorem 4.4. □

Remark 5.4 (What Theorem 5.2 does not prove). The theorem is a one-step well-posedness statement only. It does not claim that repeated application of $\mathscr{L}_\eta$ converges, that a fixed point exists, that a fixed point would be unique, or that the return map is thermodynamically optimal. It also does not address the multi-source feedback problem in which several manifest sectors update a shared ambient semigroup simultaneously.

6 Open Problems

The present note defines only a conditionally well-posed one-step return. The actual programme still leaves several substantive gaps.

1. **Constructive origin of a Type III feedback lift.** Proposition 2.6 supplies a nontrivial FD/I witness, but the general theorem still takes the source-level semigroup lift $\mathscr{F}^{[\delta\sigma]}$ as named input. A future result must derive a normal Type III lift from the calibration signal itself.

2. **Generator-level perturbation theory.** The present paper works with an exact auxiliary semigroup because it preserves complete positivity and the semigroup law cleanly. A sharper generator-level perturbation theorem on W^*-algebras, closer to the existing structural theory of dynamical semigroups on W^*-algebras [10], would have to control densely defined predual generators, domain stability, closability, and resolvent bounds in a regime where pointwise σ-weak continuity is the natural topology and norm continuity is usually unavailable. Naive first-order generator perturbations may fail to preserve complete positivity, unitality, or even semigroup well-posedness.

3. **Weakening the commutation input.** FB-C requires strict algebra-level commutation between the feedback lift and the source semigroup. A weaker theorem based on approximate commutation, asymptotic commutation, or commutation only on the relevant stable subalgebras would be more flexible and physically more natural, but it would need a replacement for the exact time-ordering argument used in Section 3. The finite witness shows explicitly that strict commutation can change decay rates while still only reweighting joint spectral sectors.

4. **Quantitative coupling thresholds.** The small-coupling window $[0, \eta_*)$ is left abstract. The programme still lacks an explicit estimate of η_* in terms of Lyapunov constants, semigroup norms, and the local BKM/Fisher geometry.

5. **Persistence of the Foundation I chain.** DF-A(1) is a real interface burden. One needs a theorem stating when persistent projections, foreground frames, and stable-hull inputs are robust under source-side feedback.

6. **A valid risk or distortion interface.** Theorem 4.4 controls only the supplied score $\mathfrak{b}_{\mathrm{loc}}$. A sampling model with an estimation-risk theorem, or a rate-distortion model with a genuine bottleneck, is required before any actual minimax deficit can be transported.

7. **Iterated perfuming and fixed points.** Once $\mathscr{L}_\eta$ is applied repeatedly, one faces an entirely new dynamical question: do the semigroups converge, cycle, or wander? The present paper does not address that long-time behavior.

8. **Thermodynamic cost of the source-level return.** T-DOME III gives model-dependent reset costs for the calibration procedure at the manifest/ego level. Foundation III has not integrated a specified irreversible implementation with the source-side semigroup perturbation.

9. **Multi-source feedback.** Foundation II handled the two-source shared-sector problem at inclusion level, but it did not address simultaneous feedback from multiple manifest sectors onto one common ambient source dynamics. Cross-source feedback remains open.

10. **Type III modular obstacles.** Even if a feedback lift is available, one still faces the usual Type III issues: modular invariance of the relevant subalgebras, possible non-uniqueness of conditional expectations, central-decomposition effects in non-factorial regimes, and the absence of a canonical bounded algebra-side generator picture that could be perturbed by finite-rank feedback data.

7 Conditional Input Ledger

The theorem chain depends on the following named inputs.

1. **FI-A: Foundation I source-to-manifest interface.** The single-source conditional chain from a source datum to a manifest pair is available in the sense restated in Appendix A.

2. **TD3-A: admissible calibration signal.** A one-step natural-gradient update signal $\delta\sigma$ exists and lies inside the T-DOME III Lyapunov neighborhood.

3. **FB-A: CP source lift.** The calibration signal admits a pointwise σ-weakly continuous semigroup lift by normal completely positive unital maps on $\mathcal{A}_U$.

4. **FB-B: small-signal differentiability.** The lifted feedback semigroup is differentiable at zero coupling so that the first-order map Γ is defined; its normalization $\mathfrak{G}_{\delta\sigma}(\mathbf{1}) = 0$ is inherited from unitality.

5. **FB-C: chain compatibility.** The feedback lift commutes with the ambient source semigroup and preserves the named structural inputs needed to rerun the chosen Foundation I chain.

6. **FB-FD: finite commuting-dephasing witness.** Proposition 2.6 verifies nontrivial $\delta\sigma$-dependence, normal UCP dynamics, strict commutation, and a source-dynamical change. It is a model check, not a Type III construction.

7. **DF-A(1): feedback-stable Foundation I chain.** The chosen Foundation I conditional inputs remain valid for $S^{[\eta]}_{\text{source}}$ on a nontrivial coupling window.

8. **DF-A(2): manifest Fisher control.** The modified manifest-side Fisher sensitivity and ego-rigidity terms satisfy the comparison inequalities controlled by $\alpha_F(\eta)$ and $\beta_{\text{ego}}(\eta)$.

9. **DF-A(3): supplied local comparison parameters.** The coefficient $\rho(\eta) < \frac{1}{2}$ and the Fisher quantities define $\mathfrak{b}^{[\eta]}_{\text{loc}}$; no Paper III reconstruction-loss theorem is inherited.

10. **Downgraded one-step return status.** The theorem-level output is a conditionally well-posed one-step return map. Its tangent is controlled only at fixed t as $\eta \downarrow 0$, or on windows $\eta t \ll 1$. Iteration, convergence, and fixed points remain open.

A Imported Interfaces

This appendix records the external results used by the body in a self-contained theorem-ready form. The statements are imported interfaces, not re-proved results.

Remark A.1 (What is and is not imported). Foundation III imports a conditional source-to-manifest interface from Foundation I and a model-level calibration interface from T-DOME III. It does not import the withdrawn Paper III reconstruction-loss theorem; only the algebraic comparison formula is retained as supplied data. Foundation II is cited in the Introduction solely to locate the later multi-source feedback problem and does not enter any proof or named input in the present paper.

Proposition A.2 (Imported Foundation I interface)**.** *Let*

$$S_{\text{source}} = (\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$$

be a source datum, where $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$ is a Type III$_1$ standard-form von Neumann algebra and $\{\mathcal{E}_t\}_{t\geq 0}$ is a pointwise σ-weakly continuous semigroup of normal completely positive unital maps on $\mathcal{A}_U$. Suppose:

1. *there exists a persistent Nakajima–Zwanzig projection $\mathcal{P}$ whose projected predual dynamics admits a memory kernel $\mathcal{K}_*(t,s)$ [8, 9, 7];*

2. *a specified finite-coordinate coding model satisfies the Foundation I rank-cost, strict-improvement, and group-action assumptions and supplies a compressed foreground frame $\mathfrak{F}$;*

3. *the foreground transfer/stable-hull construction closes so as to produce a von Neumann algebra $\mathcal{A}_c \subseteq \mathcal{A}_U$ with faithful restricted state $\omega = \langle\Omega, \cdot\,\Omega\rangle|_{\mathcal{A}_c}$;*

4. *the admissibility input needed to feed Paper I [3] is available.*

Then the conditional chain

$$S_{\text{source}} \to \mathcal{O}_A \to \mathcal{O}_B \to (\mathcal{A}_c, \omega)$$

is defined.

Remark A.3. Foundation III uses only the existence of the manifest pair produced by Proposition A.2. It does not revisit the internal stable-hull construction of Foundation I. The source-datum continuity imported here is pointwise σ-weak continuity, matching Foundation I v2. The published Foundation I v1 uses the stronger phrase "strongly continuous"; the downstream Foundation I chain used here does not depend on that stronger condition.

Proposition A.4 (Conditional T-DOME III calibration interface)**.** *Let $\mathcal{O}_B$ be a Stage B output in the sense of Foundation I. In the local calibration regime of T-DOME III there exist:*

1. *a natural-gradient update signal $\delta\sigma \in T_\sigma\Sigma$;*

2. *a Lyapunov neighborhood $\mathcal{U}_{\text{cal}}$ on which the one-step update stays controlled;*

3. *a regular prior Fisher term $\mathcal{I}_{\text{ego}} \geq 0$, interpreted as ego rigidity, for the specified local estimation model.*

These data determine the one-step manifest-side calibration step used in the body of the present paper.

Remark A.5. Proposition A.4 is a one-step interface only. It does not assert global convergence of repeated calibration updates.

Definition A.6 (Supplied local Fisher comparison score)**.** For stipulated nonnegative quantities $\mathcal{I}_F$ and $\mathcal{I}_{\text{ego}}$, and $\rho \in [0, \frac{1}{2})$, define

$$\mathfrak{b}_{\text{loc}} := \frac{(1-2\rho)\,\mathcal{I}_F}{2(\mathcal{I}_F + \mathcal{I}_{\text{ego}})}.$$

The definition makes no assertion that this score lower-bounds an actual reconstruction loss or estimation risk.

Remark A.7 (Status of the former Paper III import). The original positive drift-floor theorem is withdrawn because its admissible reconstruction class includes lossless low-dimensional orbit codes. Definition A.6 retains only the algebraic expression needed for Theorem 4.4. A future sampling model may support an estimation-risk bound, and a future bottleneck may support a rate-distortion bound, but neither result is assumed here.

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